The skin and systemic disease

The skin and systemic disease

Clues to possible systemic disease:

■

Rash ass. with joint pain, fever, weight loss, weakness,

SOB, altered bowel function

■

Rash not responding to topical tx

■

Erythema of the skin (inflammation around blood vessels)

■

Non blanching palpable purplish fixed lesion , painful and blistering >> vasculitis

■

Unusual changes in pigmentation or texture of the skin.

■

Palpable dermal lesion 2ry to granuloma , mets , lymphoma and so on…

Characteristic rash that indicates underlying systemic disease :

■

Erythema multiforme >> HSV, Hep B/C, mycoplasma pneumonia

■

Pyoderma gangrosum >> IBD , RA , hematological malignancy

■

Erythema nodosum >> IBD , streptoccocal infxn , TB , sarcoidosis , behcet's

■

Vasculitis >> Hep B /C , SLE , lymphoma , leukemia

Skin reactions ass with infections

1. Toxic erythema

■

Widespread symmetrical Maculopapular erythema (morbilliform)

■ starts on the trunk then spreads to the limbs

■

Usually asymptomatic (itchy)

■

Blanching

■

Triggered by viruses, bacteria, drugs

■

Classic presentation??

■

Tx: No need, can use;

■emollient and topical steroid if symptomatic

■treat the underlying disease

Toxic Erythema

2. EM

■

Epidemiology

■

Presentation : target lesions

■

Pathophysiology: Type IV HSR

■

Classification : EM Minor vs Major

■

Causes: according to classification, most common cause: HSV

■

Dx : Clinical

■

TX

■

Recurrence ?

■

Coarse

■

Rare associations

EM

■

The most common infectious trigger HSV

■

Systemic symptoms of the underlying infection usually precede the EM rash by 2–14 days.

EM Doll's eye Acrallyfftandstar

3 shadesDuskyred residenecrosis sometimes feet ofcolors pinkish Redagain HSV

5 14days

Erythema Nodosum

■

Most common Panniculitis.

■

Epidemiology: F >> M

■

Presentation: acute, painful nodules (Shin), especially by palpation.

■

Idiopathic (50%) vs Secondary

■

Secondary causes: Multiple, shown in the table

■

Most Common Sec : Strep

■

Skin lesions: very tender nodules(3–20 cm), not sharply marginated

■

Dx & management: Bacterial Culture, Imaging, Dermatopathology

■

Course: Spontaneous resolution occurs in 6 weeks

■

Tx: treat secondary; rest + NSAIDs (steroids ?)

Panniculitis adiposetissue inflammation

3. Erythema nodosum Tder Susa b nodules Streptococci O over theshin Yersinia multiple bilateral stestosomiasis A 20 301 idiopathic needtoexcludeany caves mostcommonlyifxn

EN: Causes

4. Erythema annulare centrifugum

Vs Tinea Corporis comitugically amenityYa Tinea pedis idiopathic gppffichongi

5. Erythema chronicum migrans

■

Caused by Borrelia burgdorferi (lyme disease)

■

Migrating erythema

■

Cutaneous inflammatory response to Borriella.

h spirochete like syphilis Thetransmitted Yarger C signbegan hematogenousspread Ws Cns Au like symptoms

Sarcoidosis

■

Unknown etiology, atypical mycobacterium may be the trigger

■

Presentation:

■

May occur with or without cutaneous disease

■

Skin lesions : Specific vs Non-specific

■

Earliest: skin-colored papules, on the face, heal without scarring.

■

Then: Plaques; annular, extremities and Buttoks, SCARRING, Chronic disease

■

Lupus Pernio: violaceous plaques on the nose and cheeks

■

EN: m/c NS skin lesion, early sarciod.

■

Scarring sarcoidosis.

Sarcoidosis CXR Bilateral hilar lymphadenopathy Noncaseating Curtibe TB Multisystemic

20 cutaneous Bigmimic

Sarcoidosis Perianths scalyptagy

Dusky infiltrated lesions: Lupus Pernio Badprognosticsign URI IRI osteolytic AcrallyThose bisons dusky violaceous color

Lupus pernio vs Rosacea midfacial erythema Sharpdermalsatientviolaceous hue

Scarring sarcoid DDx Tirifial hypertrophicpre existingscar sarcoid

Löfgren syndrome

-Erythema nodosum

-Fever

-Arthritis

-Hilar adenopathy

Acutesyndrome multisystemic Benign resolvesspontaneously

É

Diagnosis and management

■

Imaging

■

Labs

■

Lesional biopsy

■

Tx:

■

Systemic involvement: systemic steroids

■

Limited cutaneous: topical steroids, if extensive: systemic steroids, mtx, hydroxychloroquine.

Skin changes ass with hormonal imbalance

Hyperpigmentation

■

Increase in circulating hormones with melanocytestimulating activity

■

Hyperthroidism, acromegaly, Addison's disease

■

Pregnancy, OCPs -> melasma/chloasma (localized on the forehead and cheeks)

■

It may fade slowly if ultraviolet light is excluded from the affected skin using daily sun block.

Generalized Addison mulousmen localized creases melasmalyspiotsumprotection E

Hypopigmentation

■

Widespread partial loss of melanocyte function

■

Hypopituitarism (absence if MSH)

Acanthosis nigricans noBlack Insulin resistance axilla inguinal neck G Increasedthickness of epidermis obese areas extensive involvement Gastric adenocarcinoma

Necrobiosis lipoidica

■

Associated with DM

■

Not related to the severity of DM

■

Coarse not affected by controlling blood suger

■

Tx: intralesional steroids, surgery.

destructionofcollagen indermis Atrophy translucent Brs yellowishhue depositing

654Épplwith CheckHgbac

Thyroid disease

edema pitting Ohridshift I

Ettionpitting Grave's discs Generalizedmyxedema

Graves disease

■

Hyperthyroidism with diffuse goiter, Opthalmopathy and dermopathy.

■

Dermopathy: pretibial myxedema

■

Tx: steroids

Skin changes with GI and liver diseases

■

Acrodermatitis enteropathica : genetic disorder of Zinc absorption (seen in neonates)

■

Acquired zinc deficiency (AZD) occurs in older individuals due to dietary deficiency or failureof intestinal absorption of zinc (malabsorption,alcoholism, prolonged parenteral nutrition)

■

Skin changes usually appear within weeks of birth with erythematous inflamed scaly skin around the mouth, anus and eyes Zinc deficiency

Breastfedt full ATTY ofzin inthetransport ofzinc likeeczema orifices q atrally genetic to therapy minedmalabsorption notresponding

Zinc def

■

If the condition is not recognised and treated promptly then the skin can become crusted, eroded and secondarily infected.(candida, S.aureus)

■

Other findings:

■

Zinc supplementation (1mg/kg/day), should be continued until zinc levels normalise (or lifelong in inherited forms).

Vitamin C deficiency

■ occurs in those with malabsorption problems, those on a poor diet, the elderly and alcoholics

■

Vascular fragility

■

Skin lesions: Petechiae, follicular hyperkeratosis with perifollicular hemorrhage, especially on the lower legs

■

Other findings:

■

Labs: Normocytic, normochromic anemia, Serum ascorbic acid level zero. X-ray findings are diagnostic

■

Tx: Ascorbic acid imphcollagensynthesis Satb

A Hyperplastic easybleeding afterteethbrushing abnormalhair corkscrew bleeding in hair follicle

3. Pyoderma gangrenosum

■

Rapid and painful ulcerated necrotic skin areas with hypertrophic undermined purplish margins.

■

Ass with UC, Crohn’s, RA, leukemia, monoclonal gammopathy.

lookfor anycause Tibialtrauma TT papule spushde6

4rapidand_dden.O PG

4. Dermatitis herpetiformis

Buttonsdisease Celiacdisease extensor surfaces highlypruritic

5. Liver disease

Porphyria cutanea tarda according of whichenzyme ofhememetabolismis a affected theme synthesisdefect and all porphyrias have cutaneous manifested porphyrinsaccumulate

Asensitivity to sunlight t recurrentinflame cause scarring milia

Xanthomas Mefull of lipids duetogenetic disease a yellowish lipid O

Hypopigmentation disorders

■

Albinism , AR , loss of pigment of skin , eyes and hair , no pigment production

■

Piebaldism , AD , triangular hypopigmented patches , disorder of melanocyte development

■

Vitiligo localized depigmentation, sharply demarcated, symmetrical macular lesions, loss of melanocytes and melanin

■

Post- Inflammatory such as psoriasis, eczema, lichen planus and lupus erythematosus; infections, chemicals, reactions to pigmented naevi, genetic diseases

DI D

Congenital there are melanocytes defect'veTsinaseenryme eyeproblemsnystagmus foreheadforetooll Desmith'enofmelanocytes Acquired

Hyperpigmentation disorders There is wide variation in the pattern of normal pigmentation as a result of hereditary factors and exposure to the sun.

Darkening of the skin may be due:

■

1. An increase in the normal pigment melanin

■

2. Deposition of bile salts from liver disease

■

3. Iron salts (haemochromatosis)

■

4. Drugs or metallic salts from ingestion

■

5. AN is characterised by darkening and thickening of the skin of the axillae, neck, nipples and umbilicus

■

6. Post-inflammatory pigmentation is common, often after acute eczema, fixed drug eruptions and lichen planus.

■

7. In malabsorption syndromes such as pellagra and scurvy, there is commonly increased skin pigmentation.

Skin changes of underlying malignancy

Mycosis fungoides

In cutaneous Tad lymphoma differentclinical presentation

Poikeloderma (telangiectasia , reticulate pigmentation)

I Hyper Hypopigme talengectasea radiotherapy Ted tidiopathic t atrophy

Skin and Pregnancy

■

Prurigo gestationis- is a benign non-specific pruritic (itchy) papular rash that arises during pregnancy and is generally more severe in the first trimester.

■

Polymorphous eruption of pregnancy (PEP) is a pruritic erythematous rash that usually starts in the striae of the abdomen during the third trimester and can become widespread. No effect on aby

■

Pemphigoid gestationis (PG) is a rare disorder that may initially resemble PEP but develops pemphigoid-like vesicles, spreading over the abdomen and thighs (Figure 10.23). PG is an autoimmune disorder, (Small baby and higher mortality).

Eigg low weight low gestahage

PEP poly not vesicles eczematous papules duetotension Fina umbilical region no recurrence

Pemphigoid gestations

0 Buttons pemphigoid o periumbilical Baby can be affected t recurrence

GOOD LUCK